Lectio

The AI Course Curation Company

A multi-agent prototype for course alignment auditing and content creation assistance

Program	Master Applied Artificial Intelligence (MAAI)
Credits	10 ECTS — 2 semesters
Team size	2–3 students
Supervisor	Prof. Dr. Carsten Lanquillon
Co-Supervisor	Yannik Langer
Contact	carsten.lanquillon@hs-heilbronn.de
Keywords	LLM agents · multi-agent systems · RAG · AI in educational · human-in-the-loop · course design

1 · What It Is About

The Problem

University courses are not single documents — they are systems of overlapping artifacts created at different times, by different hands, without systematic cross-checking. The result is predictable: learning objectives that no longer match what gets taught, exams that probe topics the slides barely cover, module descriptions that misrepresent the actual depth of the course. Nobody checks this systematically because it would require reading everything and holding the whole course in mind at once. That is exactly what LLMs can do.

The Four Alignment Dimensions

Lectio treats alignment as a multi-dimensional problem across three core artifact groups: course metadata (module manual, syllabus, learning objectives), course content (lecture slides, notes, transcripts), and assessments (exercises, exams, quizzes). The key analytical questions arise at the boundaries between them:

- **Metadata ↔ Content — Described vs. Taught:** Is what gets taught actually what the module manual and learning objectives claim? Does the content reflect the stated scope, level, and emphasis?
- **Content ↔ Assessment — Taught vs. Tested:** Do exercises and exams probe topics that were genuinely covered in the course, and at the depth the material warrants?
- **Metadata ↔ Assessment — Intended vs. Assessed:** Do assessment tasks match the declared learning objectives and their cognitive level, for instance as specified by Bloom's taxonomy?
- **Content ↔ Delivery — Planned vs. Delivered:** Where transcripts or recordings are available: does what was actually said in the lecture reflect the prepared slides and notes, or did it diverge?

The tool does not judge which artifact is correct. It surfaces mismatches and leaves interpretation to the lecturer.

From Alignment to Creation

Where artifacts are missing or misaligned, Lectio shifts from audit to assist. It generates draft content grounded in what already exists: learning objectives derived from slides, exercises calibrated to coverage and Bloom level, a module description reconstructed from transcripts. All outputs are suggestions — the lecturer decides what to adopt, adapt, or discard. Nothing writes back to any course system automatically.

2 · The Approach: An AI Agent Company

Lectio is built on Paperclip (paperclip.ing), an open-source MIT-licensed framework for running autonomous agent organisations. Paperclip provides the organisational infrastructure out of the box: org charts, role hierarchies, ticket-based task delegation, heartbeat scheduling, per-agent budgets, and a full governance layer. This means students are not building orchestration from scratch — they are staffing a Paperclip company for the specific domain of course curation, and focusing their effort on what is novel: the course model, the alignment analysis, and the creation agents.

Goal ancestry propagation is a core Paperclip feature: every task carries the full chain of goals back to the company mission, so agents consistently know not just what to do but why. In Lectio, this means the lecturer's stated course intent flows down through the Course Director to every Division agent — grounding both analysis and generation in the same pedagogical goal.



Division 1 (Alignment Auditing) ingests uploaded artifacts, builds a structured course model through chunking, embedding, and topic clustering, then runs alignment analysis across all four dimensions described above.

Division 2 (Content Creation) uses the same course model as grounding context for generation. Drafts are always anchored to what the course actually contains, not generated in a vacuum.

The Agent Pool is explicitly pluggable. The same task can be run with different models, prompt strategies, or granularity settings, producing multiple variants. The Stakeholder agent surfaces these side-by-side.

3 · Human Oversight and Interaction Design

Lectio is designed around the principle that the lecturer stays in the loop at every step. This is not a constraint imposed on the system — it is a core architectural commitment. Paperclip supports this directly: its governance layer means agents cannot act outside their defined scope, and the human operator — here, the lecturer — can pause, override, or redirect any agent at any time.

The Stakeholder Agent as Quality Broker

The Stakeholder Management agent is the lecturer's only point of contact. It never exposes raw agent outputs directly. Instead, it:

- Translates internal findings into plain, actionable language — not technical diagnostics but concrete observations about the course
- Presents multiple variants when generation is involved, with a brief rationale for each, so the lecturer chooses rather than accepts
- Detects when Division outputs do not meet quality thresholds and requests alternatives internally — either a retry with adjusted parameters or reassignment to a different agent — before surfacing anything to the lecturer
- Feeds lecturer decisions (accept / adapt / reject) back into the company as updated constraints for subsequent tasks

Audit Logs and Behavioural Analysis

Paperclip maintains an immutable, append-only audit log of every ticket, tool call, agent decision, and inter-agent message. In most deployments this log is operational infrastructure. In Lectio it becomes a research asset.

Transparency: the lecturer can inspect the full trace of how any finding or suggestion was produced — not just the output but the reasoning chain behind it.

Research corpus: logs are the primary data source for analysing goal propagation fidelity, agent collaboration patterns, and lecturer preference trends across sessions.

Safety monitoring: the logs enable analysis of unexpected emergent behaviour — cases where agents pursue sub-goals that diverge from the course intent, or where inter-agent communication produces outputs not traceable to any single agent's instructions. The Colosseum framework (Nakamura et al., 2026) provides conceptual grounding and formal metrics for this kind of collusion detection in multi-agent systems, and serves as the analytical lens for interpreting Lectio's audit data.

Because Paperclip's logging is built in from day one, this capability is available from the first working prototype rather than retrofitted later. Semester 1 establishes the baseline log structure; Semester 2 adds the analysis tooling on top.

4 · Lectio as a Research Platform

Lectio is first and foremost a working tool. The first priority across both semesters is solid basic functionality: reliable artifact ingestion, meaningful alignment reports, and content suggestions that a real lecturer would actually use. Everything else builds on that.

Once the core is stable, the architecture is deliberately designed to be extended. The pluggable Agent Pool, full interaction logging, and the Stakeholder agent's preference tracking are not overengineering — they exist so that new ideas can be tested on a shared foundation without rebuilding from scratch. Directions that naturally follow include more capable specialist agents, systematic evaluation of model

and prompt strategies, and analysis of interaction logs to detect and understand unexpected behaviour in AI–AI and AI–human communication.

Students are not expected to exhaust these directions within the project. The base scope is well-defined and tractable for two semesters. But the architecture leaves the door open — and if a thread proves interesting during the project, there is room to go deeper. Many of these directions are equally well-suited as starting points for follow-up projects, making Lectio not just a two-semester deliverable but a lasting research platform for the group.

Project Plan

The project is structured across two semesters, with Semester 1 delivering a fully usable alignment tool and Semester 2 extending it with creation capabilities and the evaluation study.

Semester 1 — Alignment Auditing	Semester 2 — Creation & Evaluation
Artifact ingestion pipeline (all artifact types)	Division 2: content generation agents
Course model: embedding, topic clustering, metadata schema	Lecturer-initiated creation with live alignment feedback
Division 1: alignment analysis across all four dimensions	Stakeholder agent: variant comparison and preference logging
Course Director and agent communication layer	Pluggable model/prompt experiment framework
Stakeholder agent (interaction, logging, human oversight)	Evaluation study with real MAAI course materials
Lecturer dashboard and annotation interface	Emergent behaviour analysis from full interaction logs
Deliverable: working alignment tool + research report	Deliverable: full curation workbench + comparative study

Prerequisites

- Strong Python skills; hands-on experience with at least one LLM API (OpenAI, Anthropic, or local models via Ollama)
- Solid software engineering habits — reproducible, documented code is a hard requirement for an application project
- Ability to work independently and engage with the research literature
- Experience with multi-agent frameworks (LangGraph, AutoGen, CrewAI) or RAG pipelines is helpful but not required. Interest in education, AI safety, or human-AI collaboration is a genuine advantage.

What We Offer

- Close supervision with regular meetings from a team with active research interest in multi-agent AI systems and human-AI collaboration
- Access to computing resources for model inference and experimentation
- Connection to the EUonAIR Erasmus+ project - an alliance of 14 European universities working on responsible AI in higher education, giving your work an international context: www.euonair.eu

Supervisor / Contact

Supervisor	Prof. Dr. Carsten Lanquillon
Co-Supervisor	Yannik Langer
Email	carsten.lanquillon@hs-heilbronn.de
First step	Send a short email with a note on why this project interests you.

References & Further Reading

Paperclip — Open-source orchestration framework for autonomous agent companies. Org charts, goal ancestry propagation, governance, immutable audit logs. MIT-licensed.

paperclip.ing · github.com/paperclipai/paperclip

Nakamura et al. (2026) — Colosseum: Auditing Collusion in Cooperative Multi-Agent Systems. Framework for detecting and measuring collusive behaviour in LLM-based multi-agent systems using regret-based metrics and interaction trace analysis.

arxiv.org/abs/2602.15198